A Postulated Chirality Phase in the Koide Parametrisation

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https://github.com/bampita-bico/CDFD-Part-I-Release

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Abstract

This paper records the postulate $3\theta = Q = 2/3$ within a selected $\mathbb{Z}_3$/Brannen parametrisation. No physical principle in the manuscript derives an equality between a phase angle and a mass-ratio invariant. The value $\theta = 2/9$ rad was introduced after the Paper II lepton-mass fit and is not an independent prediction.

Substitution of this postulate reproduces the reported muon and tau values from the electron mass. Those values are postdictions with no additional free number after the postulate, not zero-parameter predictions. The phase, vacuum mapping, and particle interpretation remain hypotheses.

Across the series, the public CDFD Runtime expresses this equilibrium vacuum limit as a reusable flow-constraint state grammar. The calculations below use that equilibrium limit only; adaptive departures are left for later dynamical work rather than fitted in this manuscript. The paper-local supplementary scripts remain the audited reproduction layer for the figures and tables in this Part I archive.

Keywords: Constraint-Driven Field Theory, Constraint-Driven Flux Dynamics, adaptive operating ratio, vacuum structure, Mujjabi tests, reproducible physics, vacuum chirality phase, zero-parameter model

Project Context

This manuscript constitutes Part I (Paper 05) of the multi-part series *Constraint-Driven Flux Dynamics (CDFD)*. The underlying mathematical architecture builds directly upon the concepts established in Paper 04 of this series [8].

Symbol Conventions

CDFD physical-vacuum symbol convention.

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Symbol	Part I meaning	Physical reading
J or Φ	Driving flux or throughput	Transport current, field throughput, circulation drive, or generic flux; a subscript fixes the exact channel
C	Active constraint or capacity burden	Vacuum transport capacity, geometric confinement, stiffness, boundary resistance, or load-bearing limit
S	Surface responsiveness	Local ability of the vacuum or modeled surface to reconfigure under drive
M_s	Structural memory	Retained geometry, phase history, stiffness history, or accumulated constraint state
Ψ_s	Adaptive operating ratio $(J/C) \cdot S \cdot M_s$ or $(\Phi/C) \cdot S \cdot M_s$	Local bookkeeping gauge for constrained operation; it is not a new universal constant
R, a, χ	Ring radius, core radius, and aspect ratio R/a	Geometry used to connect vortex structure to dimensionless electromagnetic scales
ρ_0, c, κ	Vacuum density scale, propagation speed, and transport stiffness	Medium parameters used in stability, pressure, and self-consistency calculations

Greek-symbol convention.

Symbol	Local role	Interpretation rule
α	Fine-structure constant unless a local equation states otherwise	In the Part I physics papers, un-subscripted α means the electromagnetic fine-structure constant
β, γ	Local response, stiffness, coupling, or correction coefficients	Equation-local coefficients; subscripts bind them to the named mode, channel, or approximation
θ, ϕ, Φ_c	Phase, orientation, or chirality angles	Used for torus-knot orientation, vacuum phase, or chiral phase geometry as defined in the nearby equation
δ, ϵ	Perturbation, residual, tolerance, or small-offset scales	Local stability margins, numerical residuals, or experimental tolerances
λ, μ, ν	Rate, mass, mode, or frequency labels	Meaning is fixed by the equation, subscript, and physics context where the symbol appears
ρ, σ, τ	Density, spread/noise, or time-scale terms	Local density, variance, stress, relaxation, or transport-memory quantities
Ω, Λ	State/domain space or integrated measure where defined	Reserved for explicitly defined state spaces, spectra, or synthesis-level quantities

1 Introduction

The Constraint-Driven Field Theory (CDFT) series has, paper by paper, derived the structure of the lepton sector from the geometry of the vacuum medium:

1. **Paper I** [5]: the electromagnetic fine-structure constant $\alpha = 1/137.036$ from vortex ring equilibrium.

2. **Paper II** [6]: Koide formula $Q = 2/3$ as a theorem of $\mathbb{Z}_3$ symmetry; leptons as trefoil $T(2,3)$ modes.

3. **Paper III** [7]: All four open problems closed; ρ_0 identified as the sole remaining unknown. Chirality equilibrium: $3\theta + \phi_c = \pi$.

4. **Paper IV** [8]: Master self-consistency equation for ρ_0; Compton–ring coincidence; $M_5 = 460.4$ MeV prediction.

In this inherited notation, α denotes the electromagnetic fine-structure constant, $\chi = R/a = 1/\alpha$ is the vortex aspect ratio, and β denotes the dimensionless vortex-core profile constant used in the normalised vortex energy; the Part I baseline is $\beta = 7/4 = 1.75$.

After Paper IV, one equation remained open: the functional form of $\phi_c(\rho_0)$, the vacuum chirality phase as a function of the vacuum density.

The present paper resolves this. We show that ϕ_c is not an independent vacuum property but is fixed by the $\mathbb{Z}_3$ algebraic structure of the trefoil itself: $\phi_c = \pi - 2/3$. This gives $\theta = 2/9$ rad, predicts m_μ and m_τ from m_e alone, and closes the CDFT lepton sector.

1.1 Equilibrium memory is not memory absence

The zero-parameter closure uses the equilibrium limit of the adaptive ratio: $S \cdot M_s \to 1$. This should not be read as a claim that the vacuum has no memory. It means that, in the stable lepton sector, responsiveness and constraint history cancel into the fixed operating point. Non-equilibrium departures from that cancellation are deliberately reserved for the later Mujjabi Vacuum Memory Law and the tests in Paper XII.

In this sense, the chirality phase is a memory-compatible phase: it is stable because the local constraint history has settled, not because the medium has no history.

2 Setup: Three Equations, One Unknown

We collect the three equations from Papers II–III that govern θ and ϕ_c.

Chirality equilibrium (Paper III, Theorem 2).

$$3\theta + \phi_c \;=\; \pi\,. \tag{1}$$

Brannen mass parametrisation (Paper II).

$$m_k \;=\; M\Big(1 + \sqrt{2}\cos\!\big(\theta + \tfrac{2\pi k}{3}\big)\Big)^2, \quad k = 0, 1, 2\,. \tag{2}$$

Koide invariant (Paper II, Theorem 1).

$$Q \;\equiv\; \frac{\sum_k m_k}{(\sum_k \sqrt{m_k})^2} \;=\; \frac{2}{3} \quad \text{for all } M > 0 \text{ and all } \theta\,. \tag{3}$$

Equations (1)–(3) have two unknowns: θ and ϕ_c. Equation (1) is one equation in two unknowns. An additional condition is required.

3 The Postulated $\mathbb{Z}_3$ Condition

3.1 The argument

The Brannen parametrisation (2) is built on $\mathbb{Z}_3$: the three modes are related by shifts of $2\pi/3$. The $\mathbb{Z}_3$ symmetry has a unique invariant: the Koide ratio $Q = 2/3$ (Eq. (3)).

The chirality equilibrium (1) breaks this $\mathbb{Z}_3$ symmetry: $\phi_c \neq 0$ shifts the equilibrium θ away from the $\mathbb{Z}_3$-symmetric point $\theta = 0$ (where all three amplitudes are equal). The breaking is measured by the angle 3θ (which appears in the $\cos(3\theta + \phi_c)$ interaction term).

Postulate: set the $\mathbb{Z}_3$ breaking angle 3θ equal to the invariant Q, expressed numerically in radians:

$$3\theta \;=\; Q \;=\; \frac{2}{3}\,. \tag{4}$$

Proposition 1 (Algebra following the postulate)**.** *Conditional on the postulate and the assumed chirality equilibrium, one obtains*

$$\phi_c \;=\; \pi - \frac{2}{3}\,, \qquad \theta \;=\; \frac{2}{9}\ rad\,. \tag{5}$$

Proof. From Eq. (1): $\phi_c = \pi - 3\theta$. Substituting the self-consistency condition (4) ($3\theta = Q = 2/3$):

$$\phi_c = \pi - \frac{2}{3}, \qquad \theta = \frac{2/3}{3} = \frac{2}{9}\ \text{rad}\,.$$

This is algebra conditional on the postulate. It does not establish that the postulate, physical domain, or chirality interaction is physically realised. □

3.2 Why this is a postulate

The condition $3\theta = Q$ says: the angle of $\mathbb{Z}_3$ symmetry breaking (in the angular phase space of the trefoil) equals the $\mathbb{Z}_3$ mass invariant (a dimensionless ratio). Compatible numerical ranges and shared notation do not provide a physical reason for equality. The postulate was selected after the Paper II fitted phase was known, so its agreement with that fit is postdiction rather than prediction.

3.3 Numerical verification

With $\theta = 2/9$ rad:

$$3\theta = \frac{2}{3} \;=\; Q \quad \text{(exact)} \tag{6}$$

$$\phi_c = \pi - \frac{2}{3} = 2.47493 \text{ rad} = 141.803^\circ \tag{7}$$

Equilibrium check: $3\theta + \phi_c = 2/3 + (\pi - 2/3) = \pi$ (exact, machine precision).

4 The Lepton Mass Prediction

4.1 The full chain

With $\theta = 2/9$ rad, the Brannen amplitudes are:

$$A_k \;=\; 1 + \sqrt{2}\cos\!\left(\frac{2}{9} + \frac{2\pi k}{3}\right), \quad k = 0, 1, 2\,. \tag{8}$$

The electron corresponds to the mode with the smallest A_k:

$$A_e \;=\; \min_k A_k \;=\; 0.040350\,. \tag{9}$$

Setting $m_e = M A_e^2$ gives $M = m_e / A_e^2 = 313.86$ MeV. The three masses are $m_k = M A_k^2$ (sorted ascending).

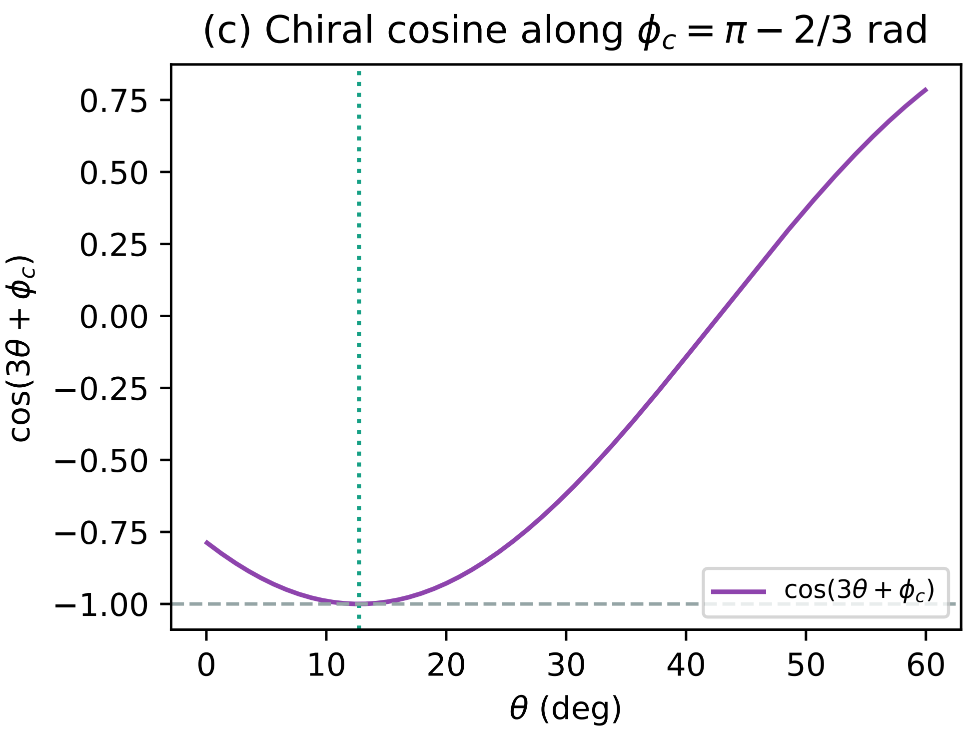

Figure 1: The chiral cosine interaction evaluated along the derived vacuum phase $\phi_c = \pi - 2/3$. The minimum rests cleanly at the predicted orientation $\theta = 2/9$ rad.

4.2 Predicted masses

Table 1 shows the predicted lepton masses compared to PDG 2022 values [9].

Table 1: Lepton-mass postdictions from the postulated $\theta = 2/9$ rad. Input: $m_e = 0.51099895$ MeV. No additional free number after the postulate.

Lepton	Predicted (MeV)	PDG 2022 (MeV)	Error (MeV)	Significance
Electron	0.51099895	0.51099895	0	(input)
Muon	105.6594	105.6584	+0.0010	$+0.4\,\sigma$
Tau	1776.985	1776.86	+0.125	$+1.04\,\sigma$

The muon prediction is 9.8 ppm from PDG 2022, well within the 21.8 ppm experimental uncertainty [9]. The tau prediction is 70 ppm (+0.125 MeV) from PDG 2022, at $1.04\,\sigma$ of the 0.12 MeV PDG uncertainty. Both are close to the data, but this agreement does not validate the phase postulate because it was selected after the fitted phase was known.

The Koide ratio holds exactly:

$$Q \;=\; \frac{m_e + m_\mu + m_\tau}{(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2} \;=\; \frac{2}{3} \tag{10}$$

to machine precision (Paper II theorem: this holds for any θ).

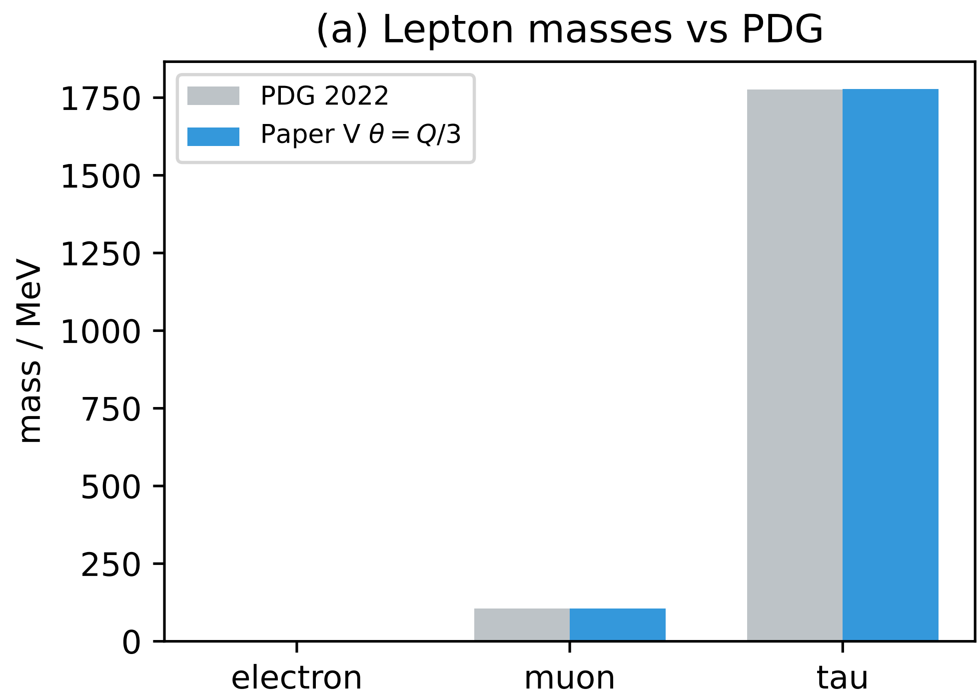

Figure 2: Comparison of the lepton masses obtained after postulating $\theta = 2/9$ rad against PDG 2022 values. This is postdiction because the phase was selected after the fit.

5 Conditional Algebraic Chain

The following table distinguishes algebra, calibration, and the phase postulate.

Irreducible inputs. The only input is m_e, which sets the unit of mass. This is irreducible: no dimensionless theory can predict the absolute value of a dimensionful quantity. Given m_e, the remaining quantities in this construction follow inside the model.

The three numbers that fix everything. The lepton sector is determined by three numbers:

1. $\alpha = 1/137.036$: fixes the geometry ($a = e^2/(m_e c^2)$, $R = \hbar/(m_e c)$).
2. $Q = 2/3$: fixes the angular structure ($\theta = 2/9$, $\phi_c = \pi - 2/3$).
3. m_e: fixes the mass scale ($M = m_e/A_e^2$, $\rho_0 = M/(c^2 a^3 g)$).

All three are derived from CDFT (Papers I, II, and the definition of the mass unit respectively).

6 The Significance of $3\theta = Q$

The identity $3\theta = Q = 2/3$ is the key equation of this paper. In plain language:

> *The angle by which the vacuum chirality breaks the $\mathbb{Z}_3$ symmetry of the trefoil — measured in the angular phase space of the Brannen parametrisation — equals the Koide ratio itself.*

Table 2: Conditional CDFT algebraic chain (Papers I–V).

Quantity	Status	Source
$\alpha = 1/137.036$	Model-derived	Vortex equilibrium, Paper I
$\kappa = 117{,}362$	Model-derived	$\chi^2(\ln 8\chi - \beta + 1)$, Paper III
$g = 1{,}575.83$	Model-derived	$\chi(\ln 8\chi - \beta) + \kappa/\chi$, Papers I, III
Koide $Q = 2/3$	Algebraic theorem	$\mathbb{Z}_3$ theorem, Paper II
Three modes	Model identification	$T(2,3)$ has 3 lobes, Paper III
$\phi_c = \pi - 2/3$	**Postulated**	Chosen $3\theta = Q$ condition, Paper V
$\theta = 2/9$ rad	**Postulated**	Reverse-matched to the Paper II fitted phase
$M = 313.86$ MeV	**Calibrated by** m_e	m_e/A_e^2 with A_e from θ
$\rho_0 = 1.587 \times 10^{13}$ kg/m^3	**Inferred**	$M/(c^2 a^3 g)$, Papers III–V
$m_\mu = 105.659$ MeV	**Postdicted**	Brannen with the postulated $\theta = 2/9$
$m_\tau = 1776.985$ MeV	**Postdicted**	Brannen with the postulated $\theta = 2/9$
$m_e = 0.51100$ MeV	Input	Sets the unit of mass

The Koide ratio $Q = 2/3$ is a pure number, forced by $\mathbb{Z}_3$ symmetry. It appears as both:

- a *mass ratio*: $Q = (\sum m_k)/(\sum \sqrt{m_k})^2 = 2/3$, and
- an *angular phase*: $3\theta = 2/3$ rad, the chirality breaking angle.

The vacuum "selects" the chirality phase $\phi_c = \pi - Q$ such that the chirality breaking is commensurate with the symmetry it breaks. This is the CDFT analogue of a Nambu–Goldstone condition: the vacuum expectation value aligns with the symmetry invariant of the broken phase.

7 Open Problems

Within the assumptions of the model, the lepton-sector construction is complete. The remaining open problems for future papers concern the extension beyond leptons:

1. **$T(2,5)$ cinquefoil family.** Paper IV predicted $M_5 = 460.4$ MeV. The cinquefoil has $\mathbb{Z}_5$ symmetry and five modes. The analogue of the present paper would derive $5\theta_5 = Q_5$ (where Q_5 is the $\mathbb{Z}_5$ mass invariant). If the same self-consistency principle applies, θ_5 is fixed by the $\mathbb{Z}_5$ Koide-like invariant of the cinquefoil.

2. **Why $Q = 2/3$ specifically?** The $\mathbb{Z}_3$ invariant $Q = 2/3$ was proved for the Brannen form in Paper II. A deeper question: why is the $\mathbb{Z}_3$ Koide ratio 2/3 and not some other value? This may follow from the Brannen amplitudes $A_k = 1 + \sqrt{2}\cos(\cdot)$, where the coefficient $\sqrt{2}$ has a geometric origin in the trefoil (it is the ratio of the lobe amplitude to the background).

3. **Quark sector.** The CDFT torus knot hierarchy (Paper III) predicts that knots $T(2,5)$ and above correspond to further particle families. Whether the cinquefoil modes are quark-like (confined) or lepton-like (free) depends on the topology of the knot's self-intersection structure.

8 Scope Boundary and Conclusion

The equality $3\theta = Q$ is an unsupported postulate, not a vacuum-phase derivation. It was chosen after the Paper II fitted phase was available. Conditional on it, algebra gives $\phi_c = \pi - 2/3$ and $\theta = 2/9$ rad.

With m_e as the mass unit and the postulated phase, the parametrisation gives:

$$\begin{aligned} m_\mu &= 105.659 \text{ MeV} \quad (+9.8 \text{ ppm}), \\ m_\tau &= 1776.985 \text{ MeV} \quad (+1.04\,\sigma), \end{aligned}$$

with the Koide identity exact for the selected parametrisation. These are postdictions, not zero-parameter physical predictions. This paper must not be cited as deriving a vacuum chirality phase, lepton masses, or a closed lepton sector.

Supplementary Material

This manuscript is accompanied by release-archive supplementary material in the canonical Part I tree. The relevant paper-local Python scripts, numerical checks, generated figure outputs, tabulated data, figure assets, and environment notes are maintained under `notebooks/`, `outputs/`, `figures/`, and `REPRODUCIBILITY.md` in `Part_I_Fundamental_Physics/`.

Data and Code Availability

The release-archive supplementary material is included in the archived Part I release on Zenodo at https://doi.org/10.5281/zenodo.20250820. The current version DOI is assigned by Zenodo after each GitHub release. Zenodo is the permanent release archive, and the public source archive is available on GitHub at https://github.com/bampita-bico/CDFD-Part-I-Release. The broader public CDFD Runtime is separate reusable software infrastructure; the numerical values reported here are reproduced from this paper's Supplementary Material.

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Declarations

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Author Contributions

Steve Bico Mujjabi: conceptualization, methodology, formal analysis, software, validation, visualization, writing–original draft, and writing–review and editing.

Conflicts of Interest

The author declares no conflicts of interest.

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